Fiscal Year:	FY 2006	**Task Last Updated:**	FY 01/08/2007
PI Name:	Tai, Yu-Chong Ph.D.		
Project Title:	Handheld Body-Fluid Analysis System for Astronaut Health Monitoring		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Technology Development Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	91125	**Congressional District:**	29
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Biomedical Research & Countermeasures 03-OBPR-04
Start Date:	04/01/2004	**End Date:**	03/31/2008
No. of Post Docs:	0	**No. of PhD Degrees:**	3
No. of PhD Candidates:	3	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Ho, Chih-Ming (University of California, Los Angeles) Kasdan, Harvey (IRIS International)		
Grant/Contract No.:	NCC 9-58-TD00406		
Performance Goal No.:			
Performance Goal Text:			
	(1) The specific aim of this project is to focus on the development and demonstration of an automated handheld blood count instrument, which is easy for astronaut to operate. A monolithic chip will be developed to perform measurements of RBC count, MCV, hematocrit, WBC count and differential. (2) We have accomplished efficient WBC/RBC separation on a new MEMS device last year. Based on the understanding of the underlining theory, we designed and tested 2nd version with improvements in various aspects. We fabricated two versions of impedance sensor for blood cell count and achieved very good signal to noise ratio. (3) Efficient separation of WBCs and RBCs is an important first step. Because of the large ratio of RBCs to WBCs, to count 200 WBCs, we need to count over 105 if we don’t do separation at first. Also cell volume for WBCs and RBCs are		

Task Description:	different, the design parameters for downstream sensing should be different to achieve high sensitivity. Now we have WBCs and RBCs separated, we will be able to count WBCs and RBCs in different components of the system and count RBCs in the order of 103. This reduction will shorten the processing time and make the sensor design part easier. The success of micro-fabricated impedance sensors enables us to do cell counting. We already demonstrated sensing and counting of bovine RBCs. With different geometrical design, impedance sensors for WBCs can be achieved. (4) For the coming year, we will work on the 1st prototype of the device, which is based on separation then count strategy. This is part of the integration, which will combine the two successful devices we have made. We will continue our work on WBC differential. Also we will investigate various sample aspiration and dilution methods. We will start work on hematocrit measurement on chip.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The devices under development can be used for earth-based applications. The proposed device use cartridge and hand held system. The cartridge will be cheap and disposable. The results will be available almost immediately to the patients without going through central lab facilities. The device can be used in emergency room, on ambulance as well as at home. As the senior population continuous to grow, this kind of device might find more and more application areas in point-of-care situations.
Task Progress:	1. According to the original proposal, we will be finishing work on RBC filter, WBC filter, RBC count and WBC count and start working on RBC MCV and WBC differential. We are on the schedule. 2. We demonstrated a new MEMS device with various improvements for highly efficient RBC/WBC separation. 3. We have made two versions electrical impedance sensors. High sensitivity and signal to noise ratio are achieved. We have demonstrated the ability for cell counting. Measurement of RBC MCV and cell count are currently under investigation. 4. We designed and tested a unique deionization device which is not in the original proposal, but we think it will be beneficial for the following years. 5. We have designed a unique direct hematocrit measurement device, which is a task supposed to be conducted in the third and fourth year. We will compare the results we got from this new device with spinning capillary tube measurement currently used in clinics.
Bibliography Type:	Description: (Last Updated: 08/30/2018)
Abstracts for Journals and Proceedings	Charlotte C Kwonga, Nan Li, Chih-Ming Ho. "Studies of deionization and impedance spectroscopy for blood analyzer." . 2005 October;. , Oct-2005
Abstracts for Journals and Proceedings	S Zheng, Y C Tai and H L Kasdan. "MEMS Device for Continuous Blood Separation." . 2005 October;. , Oct-2005
Abstracts for Journals and Proceedings	S Zheng, Y C Tai. "Streamline based design of a MEMS device for continuous blood cell separation." . Submitted for Publication, 2006 June;. , Jun-2006
Papers from Meeting Proceedings	Zheng S, Tai YC, Kasdan H. "A micro device for separation of erythrocytes and leukocytes in human blood." Annual International Conference of the IEEE Engineering in Medicine and Biology Society. Conf Proc IEEE Eng Med Biol Soc. 2005;1(1):1024-1027. PMID: 17282361 , Sep-2005